

VPPL SOLAR & WIND ENERGY SCHEDULING SYSTEM

INTELLIS INTRADAY LOGIC
ARCHITECTURE

Metadata Field	Specification Details
Document ID	DOC-M04-S01-005
Document Title	Intellis_Intraday_Logic_Architecture
Version	v1.0
Prepared by	Chaitanya Gore
Target Schedule Type	Intraday Rolling Revision Scheduling (15-Minute Grid Intervals, 96 Daily Blocks)
Target Audience	Developers, Junior Developers / Interns
Classification	Official Production Technical Specification — Complete Implementation Blueprint
Last Updated	2026-10-08
Directory Name	schedule/, schedule/modules/weather/, schedule/modules/meter/, schedule/modules/multi_generator/, schedule/modules/llm/
File Name	intellis_ai_lambda.py, intellis_ensemble_gti_ai.py, wind_ensemble.py, meter_normalizer.py, multi_generator_engine.py, strategic_arbiter.py

Document Revision History & Implementation Mandate

Revision	Release Date	Primary Author	Detailed Summary of Modifications & Architectural Scope
v1.0	2026-10-08	Chaitanya Gore	Initial Complete Production Specification. Formulates end-to-end low-level algorithmic logic for both Solar and Wind Intraday scheduling. Integrates: (1) Ineichen clear-sky ephemeris and 143-member NWP super-ensemble with physics-guided Kt spline; (2) Faiman/Sandia thermal cell convection; (3) 3-slot diurnal ensemble with agency diversity capping; (4) Real-time SCADA telemetry performance ratio horizon bridging; (5) Dynamic DynamoDB plant status change event processing (full shutdown, partial DC derates, AC export curtailment caps); (6) Multi-generator asset decomposition with column preservation across snapshot, latest, and current_final files; (7) Wind aerodynamic hardware profiling, IEC 61400-12 air density normalization, Hellmann diurnal shear, and Jensen's inequality safe non-linear power conversion; (8) AWS EventBridge 31 rolling gates orchestration; (9) OpenRouter strategic LLM arbiter with 3-tier failover matrix and 0ms deterministic physics fallback; (10) Four physical guardrails; and (11) Multi-state regulatory DSM penalty accounting and multi-path Amazon S3 enterprise data lake synchronization.

Codebase Directory & Implementation File Tree Structure

The algorithmic logic, mathematical models, and data ingestion pipelines specified within this document are implemented across the following structured module repository hierarchy under schedule/. Each component is isolated for serverless AWS Lambda execution with zero cross-site dependency.

schedule/	
├─ intellis_ai_lambda.py	[Master Lambda Entrypoint & Orchestrator]
├─ config.py	[Plant Configurations, Capacities & Thresholds]
├─ build_and_deploy_all_lambdas.py	[Multi-Plant Automated Deployer & ECR Packager]
├─ modules/	
│ └─ weather/	
│ └─ intellis_ensemble_gti_ai.py	[Solar Ineichen Clear-Sky & 143 NWP Spline]
│ └─ wind_ensemble.py	[Wind Aerodynamic, Density & OEM Power Curves]
│ └─ openmeteo_ensemble.py	[143-Member Super-Ensemble Weather Fetcher]
│ └─ satellite_virtual_meter.py	[Virtual Satellite Irradiance Fallback]
│ └─ time_features.py	[Diurnal Feature Extraction & Solar Geometry]
│ └─ meter/	
│ └─ meter_normalizer.py	[SCADA Telemetry Ingestion & Bias Blending]
│ └─ multi_generator/	
│ └─ multi_generator_engine.py	[Multi-Inverter Asset Decomposition Engine]
│ └─ ztrac_asset_schedule.py	[Multi-Gen Column Reassembly & Output Preservation]
│ └─ llm/	
│ └─ strategic_arbiter.py	[OpenRouter LLM Strategic Arbiter]
│ └─ predictor.py	[Quantile Scaling & Deterministic Physics Fallback]

Implementation File Mapping & Component Responsibilities:

Directory / Path	Code File Name	Architectura	Core Logic & Functions
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		I Role	Implemented
schedule/	intellis_ai_lambda.py	Master Handler	lambda_handler(), EventBridge event parsing, freeze boundary partitioning, DynamoDB state lock, 4 guardrails, S3 upload.
schedule/modules/weather/	intellis_ensemble_gti_ai.py	Solar Ephemeris	Ineichen clear-sky transposition, 143-member NWP fetch, Kt cubic splines, Faiman cell temperature derating.
schedule/modules/weather/	wind_ensemble.py	Wind Aerodynamics	IEC 61400-12 air density, Hellmann shear, Jensen's-safe non-linear OEM power conversion, 3-factor park derate (eta=0.89).
schedule/modules/meter/	meter_normalizer.py	SCADA Ingestion	extract_scada_wind_telemetry(), outlier trimming, recency-weighted dynamic bias correction, 3-block momentum nudge.
schedule/modules/multi_generator/	multi_generator_engine.py	Asset Decomposition	Asset configuration parsing, individual inverter MW generation, capacity scaling, sub-generator allocation.
schedule/modules/multi_generator/	ztric_asset_schedule.py	File Reassembly	Multi-generator column preservation across snapshot, latest, and current_final_schedule.csv files.
schedule/modules/llm/	strategic_arbiter.py	Strategic Arbiter	OpenRouter GPT-5.6 Luna invocation, asymmetric DSM risk regime classification, quantile scaling factor derivation.
schedule/modules/llm/	predictor.py	Fail-Safe Matrix	3-tier API key rotation, JSON validation, deterministic Oms physics baseline fallback (WIND_PHYSICS_BASELINE).
schedule/	config.py	Config Store	Plant coordinates, AC/DC nameplate capacities, DynamoDB table mappings, S3 bucket names, threshold parameters.
schedule/	build_and_deploy_all_lambdas.py	CI/CD Tooling	Parallel Docker Buildx packaging, ECR push, and zero-downtime pointer update across

			27+ site Lambdas.
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1. System Architecture & Regulatory Governance Overview

The Intellis Intraday Scheduling System is an enterprise-grade automated forecasting and grid scheduling platform designed specifically for renewable power utility assets operating within the Indian national electrical grid. The system complies strictly with the statutory mandates of the Indian Electricity Grid Code (IEGC 2023), Central Electricity Regulatory Commission (CERC) regulations, and State Electricity Regulatory Commission (SERC) frameworks, including Telangana (TSERC), Madhya Pradesh (MPERC), and Maharashtra (MERC) Deviation Settlement Mechanism (DSM) rules.

1.1 The Role of Intraday Rolling Revisions in Grid Operations

Unlike Day-Ahead (DA) scheduling, which is submitted once daily by 11:30 AM IST for the subsequent day (blocks 1 to 96), and Week-Ahead (WA) scheduling, which provides medium-range resource projections, Intraday (ID) scheduling is a continuous, rolling operational loop. Intraday scheduling enables power generators to adjust their dispatch commitments closer to real-time delivery to account for dynamic meteorological variations, cloud transients, wind speed shifts, forced equipment trips, scheduled maintenance shutdowns, and State Load Despatch Centre (SLDC) curtailment instructions.

Under CERC / SERC regulations, commercial power generators are penalized for any energy deviation between actual generation $A(b)$ and official scheduled generation $P_{\text{schedule}}(b)$ that exceeds statutory percentage deadbands (e.g., +/-10% or +/-15%). Because solar irradiance and wind velocity are highly volatile, relying solely on Day-Ahead schedules would incur catastrophic commercial DSM penalties. The Intellis Intraday engine executes at automated revision gates throughout the day, synthesizing live SCADA telemetry, multi-agency weather super-ensembles, physical plant hardware profiles, and real-time plant status change events to publish revised, legally binding 96-block dispatch schedules.

1.2 15-Minute Time Block Discretization and Grid Indexing

The Indian electrical grid operates on a standardized 15-minute time block convention. A standard calendar day from 00:00:00 IST to 24:00:00 IST is partitioned into exactly ninety-six (96) discrete dispatch time blocks, indexed b in $\{1, 2, \dots, 96\}$.

$$t_{\text{start}}(b) = (b - 1) \times 15 \text{ minutes}, t_{\text{end}}(b) = b \times 15 \text{ minutes}, t_{\text{mid}}(b) = t_{\text{start}}(b) + 7.5 \text{ minutes}$$

Where: Discretization of day into 96 blocks where b in $\{1, \dots, 96\}$.

Block Index (b)	Time Interval (IST)	Midpoint (t_mid)	Operational Classification (Solar)	Operational Classification (Wind)
1	00:00 - 00:15	00:07:30	Deep Night (GTI = 0, P = 0.00 MW)	Nocturnal Boundary Layer Jet
24	05:45 - 06:00	05:52:30	Astronomical Dawn / Solar	Nocturnal Jet / Morning

			Cut-in Threshold	Transition
25	06:00 - 06:15	06:07:30	Solar Generation Window Opens (Slot 1)	Morning Boundary Layer Breakdown
41	10:00 - 10:15	10:07:30	Solar Peak Generation Window (Slot 2)	Convective Daytime Lull Window
56	13:45 - 14:00	13:52:30	Solar Peak Generation Window (Slot 2)	Convective Daytime Lull Window
57	14:00 - 14:15	14:07:30	Solar Afternoon Decay Window (Slot 3)	Late Afternoon Thermal Breeze
74	18:15 - 18:30	18:22:30	Astronomical Dusk / Solar Window Closes	Evening Radiative Inversion Ramp
75	18:30 - 18:45	18:37:30	Night Zeroing Enforced (P = 0.00 MW)	Evening Radiative Inversion Ramp
96	23:45 - 24:00	23:52:30	Deep Night (GTI = 0, P = 0.00 MW)	Nocturnal Boundary Layer Jet

1.3 Statutory Gate Closure, Freeze Horizons, and Revision Lag (D)

A foundational principle of electrical grid dispatch is gate-closure latency. A revision submitted at execution time T corresponding to time block R cannot instantaneously alter current grid injection. Regional and State Load Despatch Centres enforce a mandatory freeze lag D (specified in 15-minute time blocks) between submission time and the first block where the revised schedule is legally permitted to take effect.

- **Madhya Pradesh (MPERC) Rule:** For plants regulated under Madhya Pradesh Electricity Regulatory Commission (MPERC) guidelines, the statutory revision freeze lag is six (6) blocks, representing exactly 90 minutes. Any revision submitted at block R takes legal effect starting at block $B_{act} = R + 6$.
- **Telangana / Maharashtra / CERC Rule:** For plants operating in Telangana (TSERC), Maharashtra (MERC), or under interstate CERC regulations, the statutory revision freeze lag is three (3) blocks, representing exactly 45 minutes. Any revision submitted at block R takes legal effect starting at block $B_{act} = R + 3$.
- **Mid-Block Gate Rule:** When an automated execution occurs mid-block (e.g., at 14:22 IST for a 14:30 gate), the system evaluates the target revision time snapshot. For example, at target time 14:30 (start of block 59), with a 45-minute lag (3 blocks: blocks 59, 60, 61), the revised schedule begins at block 62 (15:15 IST). All preceding blocks remain strictly frozen.

1.4 Dual-Horizon Topology: Immutable Past vs Dynamic Actionable Window

Every 96-block Intraday schedule is conceptually partitioned into two distinct physical domains at every revision gate R:

$$\mathcal{H}_{\text{frozen}} = \{b \mid 1 \leq b < B_{act}\}, \mathcal{H}_{\text{actionable}} = \{b \mid B_{act} \leq b \leq 96\}, \text{where } B_{act} = \min(96, R + D)$$

Where: Dual-horizon partition: past blocks are legally immutable; future blocks are updated.

- **Domain 1: Immutable Frozen History:** Blocks 1 through $B_{act} - 1$ are legally binding and physically immutable. They represent either historical energy already injected into the grid or future blocks currently trapped within the statutory gate-closure freeze zone. Under no circumstances may an intraday execution alter, overwrite, or re-bias values in H_{frozen} . The system must copy these values verbatim from the previous dispatched schedule (e.g., `current_final_schedule.csv`).
- **Domain 2: Dynamic Actionable Horizon:** Blocks B_{act} through 96 are open to revision. The core algorithmic engines, SCADA telemetry bridges, plant status change event engines, and LLM arbiters compute optimized dispatch schedules exclusively for this actionable window.
- **The 12-Block Tactical Window:** Within the actionable horizon, the immediate 12 blocks $[B_{act} .. B_{act} + 11]$, spanning a 3-hour forward operational horizon, represent the highest commercial risk window. This actionable sub-window is subjected to intensive multi-model arbitration, SCADA momentum nudging, and LLM strategic risk balancing.

1.5 End-to-End Hybrid Orchestration Pipeline

The end-to-end Intellis Intraday scheduling architecture executes as a serverless containerized pipeline on AWS Lambda. The system combines deterministic physical models, machine learning ensembling, AWS DynamoDB state tracking, and generative reasoning into a unified workflow:

- **Stage 1 — Orchestration Gate:** AWS EventBridge triggers the Intraday Lambda container at scheduled gate intervals (every 15 minutes for solar sites, and 31 rolling gates every 30 minutes from 06:00 to 21:00 IST for wind sites). Execution SLA is under 4.5 seconds for pure physics and under 15 seconds when invoking the strategic LLM arbiter.
- **Stage 2 — Telemetry Ingestion:** The system fetches raw meter and inverter telemetry via dual API endpoints or S3 dumps, cleansing timestamps, resampling to 15-minute grid intervals, and identifying data drops or communications outages.
- **Stage 3 — Plant Status Change Ingestion:** The system queries DynamoDB tables (e.g., `vedanjay-control-windows`, `plant status change events`) to extract active maintenance outages, full inverter shutdowns, partial DC derates, or SLDC AC export curtailment instructions.
- **Stage 4 — Deterministic Physics Engines:** The physical models compute high-resolution astronomical clear-sky irradiance (solar) or IEC 61400-12 air density and Hellmann boundary layer shear (wind).
- **Stage 5 — Multi-Agency NWP Super-Ensembling:** The system ingests 143 ensemble weather members from ECMWF, GFS, ICON, and GEM, fitting continuous cubic splines to clearness ratios (solar) or hub wind speeds (wind) and evaluating slot-wise portfolio champions.
- **Stage 6 — Closed-Loop Telemetry Horizon Bridge:** SCADA actuals are dynamically blended with physical forecasts using recency-weighted Performance Ratios (solar) or momentum bias vectors (wind) to establish the immediate actionable baseline.
- **Stage 7 — Control Windows & Multi-Gen Allocation:** DynamoDB plant status events and multi-generator sub-asset capacities are applied block-by-block, deriving constrained dispatch limits and preserving asset-level column breakdowns for multi-generator sites.
- **Stage 8 — Strategic LLM Arbiter:** The system constructs a dense diagnostic prompt and queries the OpenRouter API gateway (targeting `openai/gpt-5.6-luna`) with a 3-tier key failover matrix to balance asymmetric DSM risks and meteorological transitions.

- **Stage 9 — Physical Guardrails & Safety Clamping:** Four strict physical guardrails screen LLM outputs against night boundaries, clear-sky caps, overcast ceilings, and drivetrain/inverter ramp rate limits, falling back deterministically to physics baselines if anomalies are detected.
- **Stage 10 — Master Assembly & S3 Dispatch:** The 96-block master schedule is assembled by stitching frozen historical blocks with screened forward predictions, formatted into statutory multi-column CSVs, and uploaded to Amazon S3 data lake paths.

2. Part A — Intellis Intraday Solar Power Generation Scheduling System

Part A provides the exhaustive, mathematical, algorithmic, and implementation specification for the Intellis Intraday Solar Power Generation Scheduling System. The solar intraday architecture translates real-time SCADA telemetry, satellite observations, numerical weather prediction (NWP) atmospheric physics, thermal convective dissipation, active plant status events from AWS DynamoDB, and generative reasoning into grid-compliant, commercially optimized 96-block dispatch schedules.

2.1 Global Variables, Physical Constants & System Parameters

The following table defines the global system variables, physical constants, atmospheric coefficients, and operational parameters governing the solar intraday scheduling engine. All variables are strictly typed and bounded.

Variable Name	Symbol / Type	Standard Unit	Default / Valid Range	Functional Description & Context
BLOCK_COUNT	N_blocks (int)	Dimensionless	96	Number of standardized 15-minute dispatch intervals per diurnal cycle.
GRID_RESOLUTION_MINUTES	Delta_t (float)	Minutes	15.0	Temporal discretization interval for grid scheduling under Indian Grid Code.
SOLAR_CONSTANT	I_0 (float)	W/m ²	1361.2	Extraterrestrial solar irradiance at the top of Earth's atmosphere.
STC_IRRADIANCE	G_STC (float)	W/m ²	1000.0	Standard Test Conditions (STC) reference irradiance for PV module rating.
STC_CELL_TEMP	T_STC (float)	Celsius	25.0	STC reference photovoltaic cell junction temperature.
TEMP_COEFF_PMP	gamma_pmp (float)	1 / Celsius	-0.0035 to -0.0042	Photovoltaic module maximum power temperature derating coefficient.
FAIMAN_A	a_faiman (float)	W/(m ² K)	25.0 to 29.0	Faiman thermal convective heat transfer coefficient for PV module surface.

FAIMAN_B	b_faiman (float)	W/(m ³ K s)	1.2 to 1.5	Faiman thermal wind convective cooling velocity coefficient.
SANDIA_A	a_sandia (float)	Dimensionless	-3.47 to -3.56	Sandia module temperature empirical constant for open-rack ground mount.
SANDIA_B	b_sandia (float)	s / m	-0.045 to -0.052	Sandia wind convective cooling empirical exponential rate constant.
ELEVATION_CUTOFF_DEG	alpha_min (float)	Degrees	3.0	Solar elevation angle below which solar irradiance is forcibly clamped to 0.0.
SOLAR_WINDOW_START_BLOCK	b_day_start (int)	Index	24 (05:45 IST)	Earliest possible block for non-zero photovoltaic power generation.
SOLAR_WINDOW_END_BLOCK	b_day_end (int)	Index	74 (18:30 IST)	Latest possible block for non-zero photovoltaic power generation.
NWP_ENSEMBLE_MEMBERS	M (int)	Dimensionless	143	Total ensemble weather prediction members ingested across ECMWF, GFS, ICON, GEM.
MAX_MODELS_PER_AGENCY	K_agency (int)	Dimensionless	2	Agency diversity cap: maximum candidate models selected from any single NWP agency.
SLOT_CHAMPION_COUNT	K_slot (int)	Dimensionless	6	Number of diverse champion weather models selected per diurnal operational time slot.
FREEZE_LAG_BLOCK_S_DEFAULT	D_def (int)	Blocks	3 (45 minutes)	Statutory gate-closure freeze lag for CERC, TSERC (Telangana), and MERC (Maharashtra).
FREEZE_LAG_BLOCK_S_MP	D_mp (int)	Blocks	6 (90 minutes)	Statutory gate-closure freeze lag for MPERC (Madhya Pradesh) state grid regulations.
TELEMETRY_LOOKBACK_BLOCKS	N_scada (int)	Blocks	8	Number of historical 15-minute actuals evaluated for rolling performance ratio.
MAX_RAMP_FRACTION	R_max (float)	Fraction	0.10 (10% P _{ac})	Maximum allowable active power ramp rate change between adjacent 15-minute blocks.

2.2 End-to-End Solar Intraday Execution Flow

At each scheduled 15-minute grid revision gate R in $\{1, 2, \dots, 96\}$, the Intellis Solar Power Scheduling System executes the following twelve (12) deterministic computational stages in strict sequential dependency:

- **Stage 1: Ground Truth Ingestion & Telemetry Cleansing:** Ingest plant SCADA telemetry; resample to 15-minute blocks; apply satellite bypass for non-meter sites; enforce night zeroing for $b < 24$ and $b > 74$.
- **Stage 2: Ineichen Clear-Sky Solar Ephemeris Engine:** Compute astronomical ephemeris (solar zenith, azimuth, air mass AM); evaluate Ineichen clear-sky GHI, DNI, DHI; transpose to array POA using Hay-Davies model.
- **Stage 3: 143-Member NWP Ensemble & Kt Spline Engine:** Ingest 143 ensemble weather members; compute hourly clearness index $K_t(H)$; fit natural cubic splines; reconstruct continuous 15-minute GTI matrix.
- **Stage 4: Faiman/Sandia Thermal Convection & Base Power:** Simulate dynamic PV cell temperature $T_{cell}(b)$; compute thermal derate factor $\eta_{thermal}(b)$; generate 143 candidate AC power profiles clamped to P_{CAP_AC} .
- **Stage 5: 3-Slot Diurnal Portfolio Selection & Ensembling:** Partition solar day into 3 slots (Morning, Peak, Afternoon); score members against ground truth; enforce agency diversity capping (max 2/agency); compute Bayesian inverse-loss weights; cross-fade boundaries.
- **Stage 6: Real-Time SCADA Telemetry Horizon Bridge:** Extract live meter telemetry up to block R-1; compute rolling exponential Performance Ratio (PR); build 3-tier horizon bridge (0-30m 100% meter, 30-45m 50/50 blend, 45m+ weather ensemble).
- **Stage 7: DynamoDB Plant Status Change Event Engine:** Query DynamoDB for active control windows and plant status events; parse time windows; compute capacity derates for full shutdown, partial DC derates, and AC export curtailment.
- **Stage 8: Multi-Generator Asset Decomposition & Column Preservation:** For multi-generator solar plants (e.g. ZTRIC, ENRICH, SHAHA), allocate total dispatch across sub-assets proportionally; preserve individual asset columns in DataFrames.
- **Stage 9: Master Strategic Prompt Construction:** Assemble diagnostic prompt context: plant constraints, freeze zone $[R \dots R+D-1]$, atmospheric regime classification (Clear-Sky, Partly Cloudy, Overcast, Rain), diagnostic feedback, multi-model table.
- **Stage 10: OpenRouter Strategic LLM Arbiter & Key Failover:** Invoke OpenRouter API gateway (openai/gpt-5.6-luna) with 3-tier key rotation; parse JSON schema; enforce no-trip safety lockout; apply 0ms deterministic physics fallback on failure.
- **Stage 11: Physical Guardrail Screening & Master Assembly:** Screen raw predictions against 4 physical guardrails (Night zeroing, Clear-sky negative cut lockout, Overcast optical ceiling, Ramp-rate filter); stitch frozen history, LLM window, and tail.
- **Stage 12: Multi-State Regulatory Penalty Accounting & S3 Dispatch:** Compute commercial DSM deviation penalties (TSERC, MPERC, MERC); generate interactive visualization HTML; upload snapshot, latest, current_final, and penalty CSVs to Amazon S3.

2.3 Module S1: Dual-Path Ground Truth Telemetry Ingestion, Resampling & Hardware Profiling

Module Context: Ground truth telemetry is ingested dynamically to establish baseline operational status and calculate near-term performance ratios. The engine supports two distinct plant ingestion pathways:

- **Path A — Satellite-Derived Non-Meter Sites:** If `site_id` is registered in `SATELLITE_DERIVED_SITES` (e.g., CME solar assets without dedicated live SCADA loggers), live telemetry polling is completely bypassed. The engine falls back directly to high-resolution satellite-derived surface solar irradiance (CAMS / INSAT-3D) blended with day-ahead physics baselines.

- **Path B — Metered SCADA Utility Plants:** For metered utility plants, the system queries operational data loggers or S3 telemetry dumps. Timestamps are parsed, converted from UTC to Asia/Kolkata (IST), and resampled from irregular 1-minute to 5-minute telemetry intervals into discrete 15-minute energy blocks. Active power outputs across all inverter stations (N_{inv}) are summed: $P_{plant}(t) = \text{SUM}(P_{inv_i}(t))$.

$$A(b) = \frac{1}{|T_b|} \sum_{t \in T_b} \sum_{i=1}^{N_{inv}} P_{inv,i}(t), \text{ where } T_b = \{t \mid (b-1) \times 15 \leq \text{minute}(t) < b \times 15\}$$

Where: 15-minute grid block aggregation across inverter stations.

Outlier Filtering & Night Clamping: Raw meter readings are subjected to physical boundary checks. Any reading where $A(b) < 0.0$ due to auxiliary transformer reverse draw is clamped to 0.00 MW. Any reading exceeding $1.15 * P_{CAP_AC}$ is flagged as an ADC instrumentation spike and replaced with the moving average of adjacent blocks. Finally, hard astronomical night clamping is enforced: for all blocks $b < 24$ (prior to 05:45 IST) and $b > 74$ (after 18:30 IST), $A(b)$ is forced to 0.00 MW.

2.4 Module S2: Ineichen Clear-Sky Solar Ephemeris & Transposition Engine

Module Context: Clear-sky solar modeling provides the theoretical upper physical envelope for solar irradiance and PV generation. At each 15-minute midpoint $t_{mid}(b)$, the Ineichen clear-sky model evaluates extraterrestrial solar position, atmospheric optical depth, and plane-of-array (POA) transposition.

$$\cos(\theta_z) = \sin(\phi)\sin(\delta) + \cos(\phi)\cos(\delta)\cos(\omega), AM(b) = \frac{1}{\cos(\theta_z) + 0.50572 \times (96.07995 - \theta_z)^{-1.6}}$$

Where: Solar zenith angle θ_z and pressure-corrected relative optical air mass $AM(b)$.

Transposition to Array Tilt: The Hay-Davies transposition model decomposes clear-sky Global Horizontal Irradiance (GHI) into beam, circumsolar diffuse, isotropic sky diffuse, and ground albedo components to determine Global Tilted Irradiance (GTI):

$$POA_{cs}(b) = DNI_{cs}\cos(\theta) + DHI_{cs}\left[A_i \frac{\cos(\theta)}{\cos(\theta_z)} + (1 - A_i)\left(\frac{1 + \cos(\beta)}{2}\right)\right] + GHI_{cs} \cdot \rho_{albedo}\left(\frac{1 - \cos(\beta)}{2}\right)$$

Where: Hay-Davies POA transposition where β is array tilt, θ is angle of incidence, and A_i is anisotropy index.

Clear-Sky Power Envelope: The theoretical clear-sky electric generation $P_{clearsky}(b)$ is derived by applying plant DC capacity, nominal STC inverter conversion efficiency ($\eta_{inv} = 0.982$), and array wiring/dust derates ($\eta_{bos} = 0.965$):

$$P_{clearsky}(b) = \min\left(P_{CAP_AC}, P_{CAP_DC} \times \frac{POA_{cs}(b)}{1000.0} \times \eta_{inv} \times \eta_{bos}\right)$$

Where: Clear-sky generation envelope clamped strictly to AC interconnect capacity.

2.5 Module S3: 143-Member NWP Super-Ensemble & Physics-Guided Clearness (Kt) Spline Engine

Module Context: Weather predictions are ingested from the Open-Meteo Ensemble API, querying 143 distinct ensemble members across four premier meteorological agencies: ECMWF IFS (51 members), NOAA GFS (31 members), DWD ICON (40 members), and CMC GEM (21 members).

Failure Mode of Direct Linear Interpolation: Standard numerical interpolation (e.g. linear resampling of hourly GHI forecasts to 15-minute intervals) creates severe physical distortions. Because solar elevation changes non-linearly across the diurnal cycle, linear interpolation under-predicts midday peak irradiance by 8% to 15% and distorts morning and afternoon ramp gradients. The Intellis engine solves this through physics-guided Clearness Ratio (Kt) spline reconstruction.

$$K_t(H) = \frac{\text{GHI}_{\text{nwp}}(H)}{\max(10.0, \text{GHI}_{\text{clearsky}}(H))}, H \in \{0, 1, \dots, 23\}$$

Where: Atmospheric clearness index Kt computed at hourly synoptic timesteps.

A natural cubic spline $S_{Kt}(t)$ is fitted to the 24 hourly clearness ratios. The continuous spline is evaluated at the exact 15-minute midpoints $t_{\text{mid}}(b)$. Global Tilted Irradiance for candidate member m is reconstructed by modulating the high-resolution astronomical clear-sky POA curve by the spline clearness profile:

$$\text{GTI}_m(b) = S_{K_t, m}(t_{\text{mid}}(b)) \times \text{POA}_{\text{cs}}(b), \text{ for all } b \in \{1, \dots, 96\}$$

Where: Physics-guided reconstruction of 15-minute tilted irradiance matrix.

2.6 Module S4: Faiman & Sandia Convective Thermal Cell Model & Physics Base Generation

Module Context: Photovoltaic conversion efficiency degrades linearly with increasing silicon semiconductor junction temperature. The engine applies the Faiman/Sandia thermal convection model to compute dynamic cell temperatures for every member m and block b :

$$T_{\text{cell}, m}(b) = T_{\text{amb}, m}(b) + \frac{\text{GTI}_m(b)}{a_{\text{faiman}} + b_{\text{faiman}} \cdot v_{\text{wind}, m}(b)}$$

Where: Dynamic PV cell temperature incorporating ambient temperature and convective wind cooling.

Thermal Power Derating & Empirical Transfer Multipliers: The thermal efficiency derate factor $\eta_{\text{thermal}}(b)$ is evaluated relative to 25.0 C STC. Raw DC power is simulated and multiplied by site-specific empirical array transfer multipliers $M[b]$ (derived from historical seasonal calibration to capture terrain shading and soiling):

$$\eta_{\text{thermal},m}(b) = 1.0 + \gamma_{\text{pmp}}(T_{\text{cell},m}(b) - 25.0), P_{\text{base},m}(b) = \min\left(P_{\text{CAP_AC}}, P_{\text{CAP_DC}} \cdot \frac{\text{GTI}_m(b)}{1000.0} \cdot \eta_{\text{thermal},m}(b)\right)$$

Where: Simulated member generation clamped strictly to clear-sky envelope and AC capacity.

2.7 Module S5: 3-Slot Diurnal Portfolio Selection & Inverse-Loss Weighting Engine

Module Context: Diurnal solar atmospheric dynamics exhibit distinct physical regimes throughout the day. Convective cloud formation, morning haze dissipation, and afternoon monsoonal squalls affect meteorological model accuracy differently. The 96-block solar day is partitioned into three independent operational time slots:

- **Slot 1 — Morning Ramp Window:** Blocks 25 to 40 (06:00 to 10:00 IST): Morning atmospheric boundary layer heating and solar ramp-up.
- **Slot 2 — Peak Solar Window:** Blocks 41 to 56 (10:00 to 14:00 IST): Peak solar irradiation, high cell temperatures, and maximum thermal derating.
- **Slot 3 — Afternoon Decay Window:** Blocks 57 to 74 (14:00 to 18:30 IST): Afternoon solar decay, convective cloud cover, and rapid evening ramp-down.

Candidate Loss Evaluation & Agency Diversity Capping: For each member m in $\{1, \dots, 143\}$, the historical Mean Absolute Error (MAE) across recent completed days is computed for each slot. To eliminate single-agency systematic forecast collapse, an agency diversity cap is strictly enforced: no more than $K_{\text{agency}} = 2$ models may be selected from any single agency (ECMWF, GFS, ICON, GEM). The top $K_{\text{slot}} = 6$ diverse models are selected as slot champions.

$$w_{m,s} = \frac{(\text{MAE}_{m,s})^{-2}}{\sum_{k=1}^{K_{\text{slot}}} (\text{MAE}_{k,s})^{-2}}, P_{\text{slot},s}(b) = \sum_{m=1}^{K_{\text{slot}}} w_{m,s} \cdot P_{\text{base},m}(b)$$

Where: Bayesian inverse-variance loss weighting across slot champion models.

Raised-Cosine Boundary Cross-Fading: To prevent step discontinuities at slot transitions (blocks 40-41 and blocks 56-57), a 4-block raised-cosine blending filter smooths the transition:

$$\alpha(k) = \frac{1}{2} \left[1 - \cos\left(\frac{\pi(k + 0.5)}{L_{\text{blend}}}\right) \right], P_{\text{fused}}(b) = (1 - \alpha)P_{\text{slot},1}(b) + \alpha P_{\text{slot},2}(b)$$

Where: Smooth transition between consecutive diurnal slots across $L_{\text{blend}} = 4$ blocks.

2.8 Module S6: Real-Time SCADA Telemetry Ingestion & Intraday Horizon Bridge

Module Context: At revision gate R , real-time meter actuals $A(b)$ are available up to block $R-1$. However, statutory freeze lag D means the schedule cannot change until block $B_{\text{act}} = R + D$. To bridge the gap between historical ground truth and numerical weather forecasts, the engine constructs a dynamic multi-tier horizon bridge.

$$PR(b) = \frac{A(b)}{\max(0.01, P_{\text{clearsky}}(b))}, PR_{\text{recent}} = \frac{\sum_{b=R-8}^{R-1} PR(b) \cdot \exp(0.4 \cdot b)}{\sum_{b=R-8}^{R-1} \exp(0.4 \cdot b)}$$

Where: Rolling recency-weighted Performance Ratio evaluated over the last 8 daylight blocks.

Three-Tier Dispatch Horizon Bridging Formulation: For each forward block $\text{delta_b} = b - R + 1$:

- **Tier 1 — 0 to 30 min (Persistence):** For delta_b in $\{0, 1\}$ (0 to 30 minutes ahead): The schedule tracks 100% SCADA meter actuals: $P_{\text{bridge}}(b) = A(R-1)$. This immediately captures severe unforeseen contingencies such as main breaker trips or feeder isolations.
- **Tier 2 — 30 to 45 min (Hybrid Blend):** For $\text{delta_b} = 2$ (30 to 45 minutes ahead): A 50/50 hybrid blend is applied: $Kt_{\text{eff}} = 0.50 \cdot Kt_{\text{obs}} + 0.50 \cdot Kt_{\text{fctst}}$; $P_{\text{bridge}}(b) = Kt_{\text{eff}} \cdot P_{\text{clearsky}}(b)$. This smooths the handoff between local cloud transients and regional weather forecasts.
- **Tier 3 — 45 min+ (Weather Physics):** For $\text{delta_b} \geq 3$ (beyond 45 minutes ahead): The schedule transitions 100% to the pure diurnal physics ensemble: $P_{\text{bridge}}(b) = P_{\text{fused}}(b)$. Legacy midday suppression traps are completely eliminated, allowing afternoon recovery to follow radiative equations.

2.9 Module S7: DynamoDB Plant Status Change Event Engine for Solar Plants

Module Context: Plant operating status frequently changes due to inverter failures, transformer maintenance, transmission line outages, or SLDC curtailment instructions. The system queries the active AWS DynamoDB control windows table (e.g., vedanjay-control-windows, plant status change events) to ingest real-time operational constraints.

DynamoDB Event Schema & Composite Keys: Plant status events are stored with primary partition key `PLANT_CODE#EVENT_TYPE#DATE#TIME` (e.g. SHAHA#SHUTDOWN#2026-10-07#14-30, ZETRIC#CURTAILMENT#2026-10-07#14-30) and attributes: `site_id`, `asset_id` ('COMBINED' or sub-generator name like 'PRANAV', 'LOKGREENB2'), `control_type` ('SHUTDOWN', 'CURTAILMENT'), `derate_type` ('FULL', 'DC', 'AC'), `capacity_val` (float), `start_time` (ISO IST string), `end_time` (ISO IST string), and `is_active` (boolean).

Event Category	Derate Type	Capacity Value	Mathematical Action on Schedule P(b)	Action on Available Capacity AvC(b)
FULL SHUTDOWN	FULL	0.00 MW	$P(b) = 0.00 \text{ MW}$ strictly for b in $[b_{\text{start}} .. b_{\text{end}}]$	$AvC(b) = 0.00 \text{ MW}$ (Plant physically unavailable)
PARTIAL DC SHUTDOWN	DC	Delta P_{dc} (e.g. 2.0 or 5.0 MW)	$P(b) = P_{\text{base}}(b) \cdot (1.0 - \text{Delta } P_{\text{dc}} / P_{\text{dc_total}})$	$AvC(b) = AvC_{\text{base}}(b) \cdot (1.0 - \text{Delta } P_{\text{dc}} / P_{\text{dc_total}})$
AC EXPORT CURTAILMENT	AC	$P_{\text{curtail_ac}}$ (e.g. 10.0 MW)	$P(b) = \min(P_{\text{unconstrained}}(b), P_{\text{curtail_ac}})$	$AvC(b) = P_{\text{unconstrained}}(b)$ (Preserves commercial rights!)

Timestamp to Block Discretization Algorithm: Given start timestamp T_{start} and end timestamp T_{end} in IST, the block range $[b_{\text{start}} .. b_{\text{end}}]$ is resolved deterministically:

$$b_{\text{start}} = \left\lfloor \frac{\text{hour}(T_{\text{start}}) \times 60 + \text{minute}(T_{\text{start}})}{15} \right\rfloor + 1, b_{\text{end}} = \left\lfloor \frac{\text{hour}(T_{\text{end}}) \times 60 + \text{minute}(T_{\text{end}})}{15} \right\rfloor$$

Where: Exact mapping of event time window to 15-minute grid blocks $[b_{\text{start}} .. b_{\text{end}}]$.

2.10 Module S8: Multi-Generator Asset Decomposition & Multi-Column Schedule Engine

Module Context: Major utility solar installations operate as multi-generator complexes comprising multiple distinct investor assets, solar parks, or injection feeders connected behind a common grid interconnect (e.g., ZETRIC/ZTRIC, ENRICH, SHAHA). In earlier legacy implementations, Intraday schedules collapsed multi-generator plants into single aggregated columns. The modern Intellis architecture enforces strict multi-column asset preservation across all Intraday pipeline outputs.

- **Sub-Asset Level Event Resolution:** For plant SHAHA, nominal AC capacity is partitioned across sub-assets: PRANAV (15.0 MW AC), LOKGREENB2 (10.0 MW AC), etc. When an event specifies SHAHA#SHUTDOWN#2026-10-07#14-30 for asset LOKGREENB2 derating 2.0 MW DC, only the LOKGREENB2 column is derated; PRANAV remains at unconstrained generation.
- **Combined Level Proportional Allocation:** When an event targets 'COMBINED' (e.g., ZETRIC#CURTAILMENT#2026-10-07#14-30 capping export at 10.0 MW AC), the curtailment is allocated proportionally across active sub-assets according to their instantaneous generation share:

$$P_{\text{asset},i}(b) = P_{\text{total_curtailed}}(b) \times \frac{P_{\text{unconstrained},i}(b)}{\sum_{k=1}^{N_{\text{assets}}} P_{\text{unconstrained},k}(b)}$$

Where: Proportional allocation of grid curtailment ceiling across active sub-assets.

- **Multi-Column CSV Integrity:** All output CSV files (snapshot, latest, current_final) maintain canonical statutory multi-column schemas: [Block_No, From, To, TOTAL_MW, ASSET_1_MW, ASSET_2_MW, ..., AvC_MW].

2.11 Module S9: Master Strategic Prompt Construction & Layer 2 Diagnostic Assembly

Module Context: To enable high-level strategic reasoning over complex atmospheric patterns, the system synthesizes plant telemetry, weather models, and historical diagnostic feedback into an information-dense prompt for the LLM arbiter.

- **Section 1 — Plant Hardware Constraints:** Includes P_CAP_AC, P_CAP_DC, inverter station count, and active maintenance outages.
- **Section 2 — Grid Regulatory Timing:** Identifies current block R, statutory freeze lag D, frozen zone $[R .. R+D-1]$, and target 12-block window $[B_{\text{act}} .. B_{\text{act}}+11]$.
- **Section 3 — Atmospheric Classification:** Classifies atmospheric regime into CLEAR_SKY ($K_t \geq 0.80$), PARTLY_CLOUDY ($0.50 \leq K_t < 0.80$), OVERCAST ($0.25 \leq K_t < 0.50$), or RAIN ($K_t < 0.25$).
- **Section 4 — Layer 2 Diagnostic Feedback:** Embeds recent block MAE, cloud passage timing, and diagnostic directives from prior revision gates.

- **Section 5 — Multi-Model Comparison Table:** Block-by-block comparison of Clear-Sky, SCADA Telemetry Bridge, ECMWF, GFS, ICON, and GEM forecasts.
- **Section 6 — Strict JSON Schema:** Specifies target JSON schema with fields: regime, quantile_bias_factor, reasoning, and block_predictions.

2.12 Module S10: OpenRouter Strategic LLM Arbiter & Resilient Multi-Key Failover Matrix

Module Context: The strategic arbiter queries the OpenRouter API gateway targeting foundation reasoning models (openai/gpt-5.6-luna). To guarantee 99.99% system uptime and prevent execution timeouts within the AWS Lambda 60-second execution envelope, a resilient 3-tier multi-key rotation and failover matrix is enforced.

$$\mathcal{K}_{\text{pool}} = \{\text{KEY}_{\text{primary}}, \text{KEY}_{\text{secondary}}, \text{KEY}_{\text{failover}}\}, t_{\text{backoff}}(i) = 1.5 \times 2^{i-1} \text{ seconds}$$

Where: API key failover pool and exponential backoff retry on HTTP 429 or 503.

Safety Lockouts & Deterministic Fallback SLA: The engine strips markdown fences, parses JSON, and validates that output schema constraints are satisfied. The `is_trip_or_curtailment` flag is deterministically locked to False (electrical tripping is managed exclusively by Module S7). If all three API keys fail or if JSON parsing errors occur, the system triggers an immediate, 0-millisecond deterministic fallback to the Module S5 fused physics baseline `P_fused(b)`. Under no circumstance can an LLM outage halt grid schedule submission.

2.13 Module S11: Physical Guardrail Screening & Master Schedule Assembly

Module Context: Before any predicted schedule can be approved for official grid dispatch, it must pass through four (4) immutable physical guardrails that deterministically override any anomalous predictions.

- **GUARDRAIL 1 — Solar Geometry Hard Night Cutoff:** For all blocks $b < 24$ (before 05:45 IST) and $b > 74$ (after 18:30 IST): $P_{\text{schedule}}(b) = 0.00$ MW strictly. Overrides any non-zero model prediction during nocturnal hours.
- **GUARDRAIL 2 — Clear-Sky Negative Cut Lockout:** When atmospheric regime is CLEAR_SKY or $K_t(b) \geq 0.85$: model predictions cannot be depressed below the physics baseline: $P_{\text{schedule}}(b) = \max(P_{\text{llm}}(b), P_{\text{base_fused}}(b))$. Eliminates artificial under-generation penalties on cloudless days.
- **GUARDRAIL 3 — Overcast Optical Cloud Ceiling:** When atmospheric regime is OVERCAST or RAIN ($K_t(b) < 0.50$): model predictions are capped strictly at $P_{\text{max}} = \max(1.10 * P_{\text{base_fused}}(b), 0.25 * P_{\text{CAP_AC}})$. Prevents over-optimistic spikes during heavy monsoon overcast.
- **GUARDRAIL 4 — Ramp-Rate Continuity Filter:** Active power change between adjacent 15-minute blocks is limited to: $\Delta P \leq \max(0.50 \text{ MW}, 0.10 * P_{\text{CAP_AC}})$. Any prediction violating this ramp bound is clipped to preserve inverter and grid stability.

2.14 Module S12: Multi-State Regulatory DSM Penalty Engine & Enterprise Amazon S3 Dispatch

Module Context: The final stage evaluates commercial deviation settlement risks under applicable state SERC regulations and synchronizes all artifacts to the enterprise Amazon S3 data lake.

- **Telangana Regulatory Framework:** TSERC (Telangana): 15% deadband. Tier 1 (15-25% deviation: Rs 0.50/kWh); Tier 2 (25-35%: Rs 1.00/kWh); Tier 3 (>35%: Rs 1.50/kWh).
- **Madhya Pradesh Regulatory Framework:** MPERC (Madhya Pradesh): 10% deadband. Tier 1 (10-15%: Rs 0.50/kWh); Tier 2 (15-20%: Rs 0.75/kWh); Tier 3 (>20%: Rs 1.00/kWh).
- **Maharashtra Regulatory Framework:** MERC (Maharashtra): 10% deadband. Tier 1 (10-12%: Rs 0.25/kWh); Tier 2 (12-15%: Rs 0.50/kWh); Tier 3 (15-25%: Rs 0.75/kWh); Tier 4 (>25%: Rs 1.00/kWh).

$$\text{Dev}\%(b) = \frac{|A(b) - P_{\text{schedule}}(b)|}{P_{\text{CAP_AC}}} \times 100\%, \text{Penalty}(b) = \sum_{k=1}^{K_{\text{tiers}}} \max(0, \text{Dev}\%(b) - \text{Tier}_{k,\text{lower}}) \times \text{Rate}_k$$

Where: Deviation percentage and commercial DSM penalty accounting formulation.

Enterprise S3 Dispatch: Generated schedules and diagnostics are uploaded via boto3 to multi-path S3 storage:

- (1) Snapshot CSV: s3://{SCHEDULE_BUCKET}/intellis Intraday solar/{SITE}/{DATE}/schedule_from_{B}_{TS}.csv;
- (2) Cumulative Latest CSV: {DATE}_latest_schedule.csv;
- (3) Official Dispatched CSV: {DATE}_current_final_schedule.csv;
- (4) Penalty Audit CSV: {DATE}_penalty_schedule.csv;
- (5) Interactive Plotly HTML: s3://{SCHEDULE_BUCKET}/raw/vedanjay/{SITE}/{DATE}/forecasts/strategy2_evaluation_{SITE}_{DATE}.html.

3. Part B — Intellis Intraday Wind Power Generation Scheduling System

Part B provides the exhaustive, mathematical, thermodynamic, and algorithmic implementation specification for the Intellis Intraday Wind Power Generation Scheduling System. Wind power generation is governed by cubic kinetic aerodynamics, atmospheric boundary layer shear dynamics, barometric air density variations, turbine wake deficits, and asymmetric regulatory penalties. Unlike solar generation, which is strictly diurnal and ceases at night, wind generation is continuous over 24 hours (all 96 blocks) and frequently peaks during nocturnal boundary layer low-level jets.

3.1 Global Variables, Aerodynamic Constants & System Parameters

The following table defines the operational parameters, aerodynamic limits, thermodynamic constants, boundary layer shear exponents, and regulatory variables governing the wind intraday scheduling pipeline:

Parameter Name	Symbol / Type	Standard Unit	Default / Valid Range	Functional Description & Context
GAS_CONSTANT_AIR	R_spec (float)	J / (kg K)	287.05	Specific gas constant for dry atmospheric air in ideal gas density formulation.
STD_AIR_DENSITY	rho_0 (float)	kg / m ³	1.225	Standard sea-level air density specified by IEC 61400-12 power curve normalization.

AIR_DENSITY_EXPONENT	p_density (float)	Dimensionless	1/3 (0.3333)	IEC 61400-12 velocity adjustment exponent based on dynamic air density.
CUT_IN_VELOCITY	v_in (float)	m / s	3.00	Hub-height wind speed threshold below which aerodynamic rotor torque equals zero.
RATED_VELOCITY	v Rated (float)	m / s	11.50 to 12.00	Hub-height wind speed at which turbine reaches nameplate generator capacity.
CUT_OUT_VELOCITY	v_out (float)	m / s	20.00 to 25.00	High-wind emergency aerodynamic pitch-to-feather and mechanical trip threshold.
STORM_RESTART_VELOCITY	v_restart (float)	m / s	18.00	Hysteresis threshold required for turbine to resume generation following storm cutout.
PARK_EFFICIENCY_TOTAL	eta_park (float)	Dimensionless	0.890 (89.0%)	Combined three-factor park derate: wake losses (0.94) x electrical (0.97) x availability (0.98).
SHEAR_EXP_NOCTURNAL	alpha_night (float)	Dimensionless	0.28 to 0.35	Hellmann shear exponent for stable nocturnal boundary layer low-level jet.
SHEAR_EXP_TRANSITION	alpha_trans (float)	Dimensionless	0.18 to 0.22	Hellmann shear exponent during morning boundary layer convective breakdown.
SHEAR_EXP_CONVECTIVE	alpha_day (float)	Dimensionless	0.10 to 0.14	Hellmann shear exponent during strong daytime convective solar heating.
SHEAR_EXP_EVENING	alpha_eve (float)	Dimensionless	0.20 to 0.26	Hellmann shear exponent during evening boundary layer radiative stabilization.
CUBIC_CURVE_EXPONENT	k_power (float)	Dimensionless	2.80	Empirical polynomial exponent for sub-rated wind turbine aerodynamic power curve.
MOMENTUM_NUDGE_WEIGHTS	w_mom (vector)	Dimensionless	[0.70, 0.45, 0.20]	Three-block exponential momentum nudge weights applied to actionable horizon blocks.
BIAS_DECAY_CONSTANT	tau_bias (float)	Blocks	8.0	Exponential decay lookback window for closed-loop SCADA telemetry bias evaluation.

MAX_DRIVETRAIN_RAMP	Delta_P_max (float)	MW / block	max(1.0, 0.15*P_rated)	Maximum allowable active power ramp delta to protect gearbox and drivetrain integrity.
EVENTBRIDGE_GATES_COUNT	N_gates (int)	Gates / Day	31	Number of automated rolling execution gates triggered daily (06:00 to 21:00 IST).
CERC_DSM_TOLERANCE	Tol_cerc (float)	Percentage	+/- 10.0% to 15.0%	Statutory allowable deviation bandwidth under CERC DSM regulations.

3.2 End-to-End Wind Intraday Execution Flow

At each 30-minute automated AWS EventBridge execution gate R (31 daily gates from 06:00 to 21:00 IST), the Intellis Wind Power Scheduling System executes ten (10) deterministic computational modules in strict sequential dependency:

- **Module W1: Plant Hardware Profiling & Aerodynamic Derate:** Ingest turbine hardware profiles (WindTurbineProfile) for Chandawasa, JGBPL, and Jewli; establish hub heights, IEC Class bounds, and 3-factor park derates.
- **Module W2: Thermodynamic Air Density & Kinetic Flux Engine:** Evaluate surface barometric pressure P_surf and temperature T_2m; compute air density rho(b); normalize wind speed via IEC 61400-12.
- **Module W3: 143-Member NWP Ensemble & Boundary Layer Shear:** Ingest 143 ensemble weather members across ECMWF, ICON, GEFS, GEM; apply diurnal Hellmann power-law shear scaling; interpolate via continuous cubic splines.
- **Module W4: Diurnal Wind Regime Slot Selection & Ensembling:** Partition 24-hour day into 4 diurnal wind regimes (Nocturnal Jet, Morning Decay, Daytime Lull, Evening Ramp); score members against ground truth; enforce agency diversity capping.
- **Module W5: Real-Time SCADA Telemetry & Closed-Loop Bias:** Ingest live SCADA actuals; apply 10% velocity outlier filter; compute recency-decayed bias error; inject 3-block momentum nudge vector [0.70, 0.45, 0.20].
- **Module W6: Jensen's-Inequality-Safe Power Transformation:** Enforce Jensen's inequality proof ($E[v^3] \geq (E[v])^3$); pass EACH candidate wind speed through OEM power curve BEFORE ensembling; apply park derate and site multipliers.
- **Module W7: AWS EventBridge & Statutory Freeze Boundary:** Orchestrate 31 daily rolling execution gates; enforce 90-min lag (MP) or 45-min lag (CERC); freeze immutable past blocks; slice 12-block actionable horizon.
- **Module W8: DynamoDB Plant Status Change Engine for Wind:** Query DynamoDB for wind curtailment and turbine trip events; apply turbine-level availability derating $N_{\text{available}} / N_{\text{total}}$; cap AC export limits.
- **Module W9: Wind LLM Strategic Arbiter & Asymmetric DSM Risk:** Classify wind regime (Steady High Wind, Diurnal Breeze, Gusty Turbulent, Low Wind Cut-in Risk); apply asymmetric CERC shortfall penalty logic; invoke OpenRouter; apply fallback SLA.

- **Module W10: 96-Block Master Assembly & Multi-Path S3 Dispatch:** Verify 24-hour continuous wind operation (NO night zeroing); format canonical 7-column master CSV; upload snapshots, cumulative latest, current final, and penalty CSVs to S3.

3.3 Module W1: Plant Hardware Profiling & Aerodynamic Derate Engine

Module Context: Wind power conversion efficiency is fundamentally governed by wind turbine hardware parameters, aerodynamic operating thresholds, and park-level aerodynamic wake interference. The hardware profile registry manages operational constants across the fleet:

Plant Name	Location / State	Total Nameplate Capacity	Turbine Model / OEM	Hub Height (z_hub)	Rotor Diameter (D_rotor)	Turbine Count
CHANDAWASA	Madhya Pradesh	15.0 MW	Suzlon S97 Class IIIA	120.0 m	97.0 m	7 x 2.1 MW
JGBPL	Madhya Pradesh	30.0 MW	Gamesa G97 Class IIIA	104.0 m	97.0 m	15 x 2.0 MW
JEWLI	Maharashtra	25.2 MW	Suzlon S111 Class S	120.0 m	111.0 m	12 x 2.1 MW

Aerodynamic Operating Velocity Limits: Turbines operate within strictly defined wind velocity thresholds: Cut-in velocity $v_{in} = 3.00$ m/s; Rated velocity $v_{rated} = 11.50$ m/s; Cut-out velocity $v_{out} = 20.00$ m/s. During severe atmospheric squalls where v_{hub} exceeds v_{out} , turbines feather blades and trip. Once tripped, generation remains locked at 0.00 MW until wind speeds subside below $v_{restart} = 18.00$ m/s for at least two consecutive blocks (30 minutes).

Combined Three-Factor Park Derate Formulation: Total park generation is significantly lower than the sum of isolated turbines due to wake aerodynamic turbulence, internal electrical collection losses, and scheduled availability outages:

$$\eta_{park} = \eta_{wake} \times \eta_{electrical} \times \eta_{availability} = 0.940 \times 0.970 \times 0.980 = 0.893 \approx 0.890 \text{ (89.0\% Total Park)}$$

Where: Three-factor park aerodynamic and electrical efficiency derate.

3.4 Module W2: Thermodynamic Air Density & Kinetic Wind Flux Engine

Module Context: Aerodynamic rotor lift and mechanical torque are directly proportional to atmospheric air density ρ . Standard OEM turbine power curves assume sea-level air density $\rho_0 = 1.225$ kg/m³. Because inland Indian wind farms (such as Chandawasa and JGBPL in Madhya Pradesh) sit at elevated plateaus (450m to 600m above sea level) with summer ambient temperatures exceeding 42 C, actual air density drops to 1.08 - 1.14 kg/m³. Failing to correct for density causes severe over-prediction errors of 10% to 15%.

$$\rho(b) = \frac{P_{surf}(b) \times 100.0}{R_{spec} \times (T_{2m}(b) + 273.15)}, v_{eff}(b) = v_{hub}(b) \times \left(\frac{\rho(b)}{\rho_0} \right)^{1/3}$$

Where: Ideal gas air density and IEC 61400-12 density-normalized effective wind speed.

A natural cubic spline interpolates hourly air density forecasts to continuous 15-minute midpoint intervals across all 96 blocks.

3.5 Module W3: 143-Member NWP Super-Ensemble & Atmospheric Boundary Layer Shear Engine

Module Context: Standard NWP weather forecasts provide wind speeds at standard 10-meter or 100-meter reference heights. Because modern utility turbines operate at hub heights of 104 meters to 120 meters, wind velocity must be extrapolated vertically through the atmospheric boundary layer using the Hellmann power-law wind shear formulation:

$$v_{\text{hub}}(h) = v_{\text{ref}}(h) \times \left(\frac{z_{\text{hub}}}{z_{\text{ref}}} \right)^{\alpha(h)}$$

Where: Hellmann power-law vertical wind shear extrapolation.

Diurnal Atmospheric Boundary Layer Shear Regimes: The shear exponent $\alpha(h)$ is not static; it fluctuates dramatically over the diurnal thermodynamic heating cycle:

- **Nocturnal Stable Regime (alpha = 0.28 - 0.35):** Regime 1 — Nocturnal Low-Level Jet (21:00 to 06:00 IST): Under strong ground radiative cooling, atmospheric stratification is highly stable. Vertical turbulent mixing is decoupled, producing strong nocturnal low-level jets with elevated shear: α_{night} in [0.28, 0.35]. Hub-height wind speeds frequently exceed surface anemometer speeds by 40% to 60%.
- **Morning Transition Regime (alpha = 0.18 - 0.22):** Regime 2 — Morning Boundary Layer Breakdown (06:00 to 10:00 IST): Solar ground heating initiates convective thermals, eroding the nocturnal inversion and triggering rapid downward momentum mixing: α_{trans} in [0.18, 0.22].
- **Daytime Convective Regime (alpha = 0.10 - 0.14):** Regime 3 — Convective Daytime Lull (10:00 to 17:00 IST): Intense solar insolation drives deep vertical atmospheric mixing. The boundary layer becomes nearly adiabatic and well-mixed, collapsing vertical shear: α_{day} in [0.10, 0.14].
- **Evening Inversion Regime (alpha = 0.20 - 0.26):** Regime 4 — Evening Radiative Stabilization (17:00 to 21:00 IST): Surface cooling terminates convective plumes, re-establishing thermal stratification and accelerating hub-height wind speeds: α_{eve} in [0.20, 0.26].

3.6 Module W4: Diurnal Wind Regime Slot Selection & Champion Portfolio Ensembling

Module Context: Just as in solar scheduling, wind forecast accuracy varies widely across meteorological agencies depending on the boundary layer stability regime. The 96 blocks are partitioned into four diurnal wind regimes:

- **Slot 1 — Nocturnal Jet Window:** Blocks 1 to 24 (00:00 to 06:00 IST) and Blocks 85 to 96 (21:00 to 24:00 IST): Stable nocturnal low-level jet.
- **Slot 2 — Morning Boundary Decay Window:** Blocks 25 to 40 (06:00 to 10:00 IST): Inversion erosion and downward momentum flux.

- **Slot 3 — Daytime Convective Lull Window:** Blocks 41 to 68 (10:00 to 17:00 IST): Thermal convective mixing and wind speed minimum.
- **Slot 4 — Evening Stabilization Ramp Window:** Blocks 69 to 84 (17:00 to 21:00 IST): Nocturnal stabilization and evening wind ramp.

Champion Selection: For each slot, candidate ensemble members are evaluated against historical ground truth MAE. Agency diversity capping restricts selection to a maximum of $K_{\text{agency}} = 2$ models per NWP agency. Bayesian inverse-loss weights are computed, and 4-block raised-cosine blending smooths transitions between regimes.

3.7 Module W5: Real-Time SCADA Telemetry Ingestion & Closed-Loop Bias Blending

Module Context: At revision gate R, real-time turbine SCADA telemetry (active generation P_{meas} and anemometer wind speeds v_{meas}) is ingested. Any telemetry showing velocity drops exceeding 10% per second or unrealistic power spikes is filtered.

$$\Delta_{\text{bias}} = \frac{\sum_{b=R-8}^{R-1} (A(b) - P_{\text{base}}(b)) \cdot \exp(0.4 \cdot b)}{\sum_{b=R-8}^{R-1} \exp(0.4 \cdot b)}, w_{\text{momentum}} = \begin{bmatrix} 0.70 \\ 0.45 \\ 0.20 \end{bmatrix}$$

Where: Exponentially recency-decayed bias error and 3-block momentum nudge vector.

Application to Actionable Horizon: The momentum nudge vector is applied to the first three blocks of the actionable horizon $[B_{\text{act}}, B_{\text{act}+1}, B_{\text{act}+2}]$:

$$P_{\text{live}}(B_{\text{act}} + k) = P_{\text{base}}(B_{\text{act}} + k) + w_{\text{momentum}}[k] \times \Delta_{\text{bias}}, k \in \{0,1,2\}$$

Where: Momentum nudging applied to actionable dispatch blocks.

3.8 Module W6: Jensen's-Inequality-Safe Power Transformation Engine

Module Context: Wind power is fundamentally non-linear and cubic in wind speed: P proportional to v^3 . A critical mathematical vulnerability in naive forecasting systems is averaging wind speeds across ensemble members BEFORE passing them through the turbine power curve. By Jensen's Inequality, for any strictly convex function $f(x)$:

$$\mathbb{E}[f(v)] \geq f(\mathbb{E}[v]) \Rightarrow \frac{1}{M} \sum_{m=1}^M v_m^3 \geq \left(\frac{1}{M} \sum_{m=1}^M v_m \right)^3$$

Where: Mathematical proof of Jensen's Inequality for cubic power conversion.

Severe Under-Prediction Penalty: In the steep sub-rated operating zone (5.0 m/s to 9.5 m/s), passing an ensemble-mean wind speed through the power curve underestimates expected electrical power generation by 15% to 28%! The Intellis engine strictly enforces Member-by-Member Non-Linear Power Transformation:

EACH candidate member wind velocity $v_{\text{eff},m}(b)$ is individually transformed through the OEM turbine power curve table BEFORE any ensemble averaging or ensembling occurs.

$$P_{\text{gross},m}(b) = f_{\text{OEM}}(v_{\text{eff},m}(b), \rho(b)), P_{\text{net},m}(b) = \min(P_{\text{rated}}, P_{\text{gross},m}(b) \times \eta_{\text{park}} \times M[b])$$

Where: Individual member power curve evaluation followed by park derating and site empirical multipliers.

3.9 Module W7: AWS EventBridge Orchestration & Statutory Freeze Boundary Partitioning

Module Context: Automated wind scheduling operates across thirty-one (31) daily rolling execution gates triggered via dual AWS EventBridge cron rules: `cron(30 0-15 * * ? *)` and `cron(0 1-15 * * ? *)`, executing every 30 minutes from 06:00 to 21:00 IST.

- **Statutory Freeze Lag Enforcement:** Freeze boundary calculation: $B_{\text{act}} = \min(96, B_{\text{snap}} + D)$, where $D = 6$ blocks (90 min) for MPERC (Chandawasa, JGBPL) and $D = 3$ blocks (45 min) for MERC / CERC (Jewli).
- **Immutable History Preservation:** Preserves all blocks $b < B_{\text{act}}$ verbatim from the previous `current_final_schedule.csv` to protect legal grid commitments.
- **Tactical Window Slicing:** Extracts the 12-block tactical horizon $[B_{\text{act}} .. B_{\text{act}}+11]$ (3 hours) for intensive LLM strategic arbitration.

3.10 Module W8: DynamoDB Plant Status Change Event Engine for Wind Plants

Module Context: Wind farms experience turbine-level outages due to gearbox inspections, yaw mechanism faults, pitch hydraulic trips, or feeder isolations, as well as grid-directed AC curtailments. The engine queries DynamoDB for active control windows:

- **Turbine Trip Availability Derate:** When N_{down} turbines trip out of N_{total} installed turbines, the available capacity and gross generation are derated by the exact turbine availability ratio: $P_{\text{derated}}(b) = P_{\text{raw}}(b) * (N_{\text{total}} - N_{\text{down}}) / N_{\text{total}}$.
- **Grid AC Export Curtailment:** When SLDC issues an AC export ceiling (e.g. Max 10.0 MW AC), the schedule is capped at 10.0 MW: $P_{\text{schedule}}(b) = \min(P_{\text{derated}}(b), 10.0 \text{ MW})$. Crucially, the Available Capacity column `AvC_MW` preserves the full unconstrained aerodynamic capability $P_{\text{derated}}(b)$ to safeguard the generator's commercial deemed generation compensation rights.

3.11 Module W9: Wind LLM Strategic Arbiter & Asymmetric DSM Risk Engine

Module Context: Wind forecasting errors carry highly asymmetric commercial penalties under Indian DSM regulations. Under-generation shortfall (scheduling 20 MW but generating only 12 MW) forces the grid operator to draw high-cost frequency balancing power, triggering punitive deviation surcharges up to 200% of PPA tariff. Over-generation (generating more than scheduled) carries near-zero penalties or earns energy credits. Therefore, rational scheduling demands conservative bias positioning during volatile conditions.

- **Regime 1 — STEADY_HIGH_WIND:** Steady high wind (7.5 to 12.0 m/s): full rotor torque, quantile bias 1.00 - 1.05.
- **Regime 2 — DIURNAL_THERMAL_BREEZE:** Daytime thermal convection, mild lull: quantile bias 0.95 - 1.00.

- **Regime 3 — GUSTY_TURBULENT:** High vertical shear variance, gust factor > 1.35: quantile bias 0.90 - 0.95.
- **Regime 4 — LOW_WIND_CUTIN_RISK:** Speeds between 2.8 and 4.0 m/s near 3.0 m/s cut-in threshold: extreme risk of zero generation. Apply aggressive conservative quantile 0.80 - 0.90.

Jewli Explicit Safety Lockout: To ensure strict deterministic compliance, the engine executes `_is_llm_disabled_for_plant('JEWLI')`. Unless explicitly overridden via `USE_LLM_JEWLI=true`, the Jewli plant completely bypasses the LLM arbiter and deploys pure physics.

3.12 Module W10: 96-Block Master Assembly, CERC Guardrails & Multi-Path Amazon S3 Dispatch

Module Context: The final wind module enforces physical grid constraints and statutory schemas across all 96 blocks:

- **Rule 1 — Continuous 24-Hour Operation:** Strictly NO Solar Night Zeroing: Under no circumstances may nocturnal wind generation be forced to 0.00 MW. Continuous 24-hour generation is preserved across all 96 blocks.
- **Rule 2 — Drivetrain Ramp Smoothing:** Active power transitions between consecutive blocks are filtered to $\Delta P \leq \max(1.0 \text{ MW}, 0.15 * P_{\text{rated}})$ to prevent drivetrain stress.
- **Rule 3 — Capacity Bounds:** Schedule values are clamped within physical nameplate bounds: $0.00 \text{ MW} \leq P_{\text{schedule}(b)} \leq P_{\text{rated}}$.
- **Rule 4 — Canonical 7-Column Schema:** Master CSVs are constructed with canonical 7 columns: [Block, Time Interval, wind_speed_hub_m_s, air_density_kg_m3, intellis_gti, intellis_mw, schedule_mw].
- **Rule 5 — Multi-Path Enterprise S3 Synchronization:** Files are uploaded to multi-path S3 storage: snapshot CSV, cumulative latest CSV, current final CSV, and penalty CSV at `s3://{SCHEDULE_BUCKET}/intellis Intraday wind/{SITE}/{DATE}/` and `s3://{SCHEDULE_BUCKET}/raw/vedanjay/{SITE}/{DATE}/forecasts/`.

4. Part C — Integration, Artifact Schema, Edge Cases & Developer Implementation Guide

Part C synthesizes the full operational architecture into a concrete engineering reference. It provides the unified data dictionary, AWS Lambda runtime execution envelopes, a comprehensive step-by-step developer and intern implementation guide, and an exhaustive matrix of twelve (12) real-world edge cases and automated failure mitigations.

4.1 Unified Intraday Data Dictionary & Schema Specification

Every file, database record, and in-memory data frame produced or consumed by the Intellis Intraday system conforms to the following standardized schemas:

Artifact / Object Name	Storage Location / Format	Key Columns / Schema Fields	Update Cadence & Retention
SNAPSHOT SCHEDULE CSV	S3: intellis Intraday {solar wind}/{SITE}/{DATE}/s	Block_No, From, To, TOTAL_MW, {ASSET_COLS},	Every 15 or 30 min revision gate.

	chedule_from_{B}_{TS}.csv	AvC_MW	Retained indefinitely.
CUMULATIVE LATEST CSV	S3: .../{DATE}_latest_schedule.csv	Same as snapshot. Merges latest revised blocks over previous gates.	Overwritten at each revision gate.
CURRENT FINAL CSV	S3: .../{DATE}_current_final_schedule.csv	Canonical regulatory dispatch schedule submitted to SLDC / QCA.	Overwritten at each gate. Serves as official billing baseline.
PENALTY SCHEDULE CSV	S3: .../{DATE}_penalty_schedule.csv	Block, Actual_MW, Scheduled_MW, Dev_MW, Dev_Pct, Penalty_INR	Computed post-facto as actual SCADA becomes available.
DYNAMODB CONTROL WINDOWS	DynamoDB: vedanjay-control-windows	PK: PLANT#EVENT#DATE#TIME, site_id, asset_id, control_type, derate_type, capacity_val, start_time, end_time, is_active	Queried at start of every revision gate.
LLM ADVICE JSON	Local / S3: .../llm_schedule_{run_date}_block_{R}.json	regime, quantile_bias_factor, preferred_agency, is_trip_or_curtailement, reasoning, block_predictions {b: MW}	Archived per revision gate for diagnostic auditing.
EVALUATION PLOT HTML	S3: .../raw/vedanjay/{SITE}/{DATE}/forecasts/strategy2_evaluation_{SITE}_{DATE}.html	Interactive Plotly HTML containing Actual, Scheduled, ClearSky, Ensemble traces and DSM bands	Generated at each gate for operational portal viewing.

4.2 AWS Lambda Container Execution SLA & Runtime Boundaries

The production Intraday engine runs as an automated containerized AWS Lambda function deployed via Amazon Elastic Container Registry (ECR) on Python 3.11 x86_64 architecture:

- **Memory Allocation:** Configured at 2048 MB to allow fast vector operations with numpy, pandas, and scipy.
- **Execution SLA & Timeout:** Hard Lambda timeout set to 60 seconds. The pure physics execution path completes within 3.5 to 4.2 seconds; the OpenRouter LLM call path completes within 8.0 to 14.5 seconds.
- **Ephemeral Disk Space:** Mounted at /tmp with 1024 MB ephemeral storage for intermediate CSV and JSON caching.
- **IAM Policy Constraints:** Least-privilege role with permissions for dynamodb:Query, dynamodb:Scan, s3:GetObject, s3:PutObject, and logs:PutLogEvents.

4.3 Step-by-Step Developer & Intern Implementation Guide

To enable an engineering intern or new developer to construct the complete operational Intellis Intraday codebase from scratch, the implementation must follow this precise structural hierarchy and execution sequence:

- **Step 1: Codebase Directory Layout:** Establish the modular directory structure under schedule/: modules/weather/ (solar and wind NWP fetchers, splines, ephemeris), modules/control_windows/

(DynamoDB event parsers, capacity derates), modules/multi_generator/ (asset column allocators), modules/llm/ (OpenRouter client, multi-key rotation, prompt builders), and modules/s3_dispatch/ (CSV formatters, boto3 sync).

- **Step 2: Implement Ephemeris & Atmospheric Engines:** Implement solar ephemeris (Ineichen clear-sky GHI/DNI/DHI, Hay-Davies transposition) and wind thermodynamic equations (ideal gas density rho, IEC 61400-12 normalization, Hellmann shear exponent). Validate against pvlib and IEC standards.
- **Step 3: Implement 143-Member NWP Fetcher & Splines:** Construct Open-Meteo ensemble API connectors. Implement hourly clearness ratio $K_t(H)$ and wind speed cubic splines. Verify that linear interpolation is strictly avoided.
- **Step 4: Implement Thermal & Aerodynamic Conversion:** Implement Faiman/Sandia PV cell temperature models for solar. Implement member-by-member OEM power curve lookup with cubic fallback for wind (enforcing Jensen's inequality).
- **Step 5: Build Diurnal Portfolio Selection & Ensembling:** Partition the day into diurnal slots (3 slots for solar, 4 regimes for wind). Implement historical loss scoring, agency diversity capping (max 2/agency), and Bayesian inverse-loss weighting.
- **Step 6: Implement SCADA Telemetry Horizon Bridge:** Build the closed-loop telemetry bridge: rolling exponential Performance Ratio for solar and momentum nudge vector for wind. Verify smooth handoff between meter persistence and weather physics.
- **Step 7: Implement DynamoDB Control Windows & Multi-Gen Engine:** Connect to DynamoDB. Parse start/end timestamps to 15-minute blocks. Implement derating logic for FULL shutdown, partial DC derates, and AC export curtailment. Ensure multi-generator asset columns are fully preserved.
- **Step 8: Construct Strategic Prompt & OpenRouter Gateway:** Build the diagnostic prompt builder. Implement the multi-key OpenRouter caller with exponential backoff on HTTP 429/503. Implement 0ms deterministic physics fallback.
- **Step 9: Enforce Four Physical Guardrails:** Implement Guardrail 1 (solar night zeroing / continuous wind), Guardrail 2 (clear-sky negative cut lockout), Guardrail 3 (overcast ceiling), and Guardrail 4 (ramp rate filter).
- **Step 10: Assemble Master CSVs & Synchronize S3 Data Lake:** Format outputs into canonical multi-column CSVs. Upload to snapshot, cumulative latest, current final, and penalty paths on Amazon S3. Write unit tests to verify 96-block continuous completeness.

4.4 Comprehensive Edge Cases & Automated Failure Mitigation Matrix

The following matrix documents twelve (12) critical operational edge cases encountered in commercial grid scheduling, along with their automated detection triggers, algorithmic responses, and fail-safe recovery procedures:

Edge Case ID	Trigger Condition & Anomaly Description	Immediate Algorithmic Response	Fail-Safe Recovery & Fallback Mechanism
EC-01	OpenRouter API Key Exhaustion (HTTP 429 / 503 across all 3 keys).	Trigger exponential backoff; rotate through primary, secondary, and failover keys.	Instant 0ms fallback to Module S5 / W4 fused physics ensemble. Submit schedule without delay.
EC-02	SCADA Telemetry Loss / API Downtime (No actuals received for > 60 min).	Flag telemetry drop in diagnostic payload. Set SCADA confidence weight to 0.0.	Bypass live horizon bridge; rely 100% on pure weather physics ensemble.

EC-03	Extreme Solar Night Prediction (LLM predicts non-zero MW at night).	Guardrail 1 intercepts predictions in blocks $b < 24$ and $b > 74$.	Forcibly clamp $P_schedule(b) = 0.00$ MW. Log security warning.
EC-04	Wind High-Speed Storm Trip (Hub wind speed exceeds $v_out = 20.0$ m/s).	Aero engine detects cut-out exceedance; trips turbine power to 0.00 MW.	Maintain 0.00 MW until wind drops below $v_restart = 18.0$ m/s for 2 consecutive blocks.
EC-05	Conflicting Plant Status Events (Overlapping shutdown and curtailment in DDB).	Module S7/W8 identifies time overlap across active DynamoDB records.	Enforce conservative dominance: Full Shutdown (0 MW) > Partial Derate > AC Export Cap.
EC-06	Mid-Block Execution Timing (Gate triggered at 14:22 IST instead of 14:30).	Engine resolves nearest upcoming gate boundary (14:30, start of block 59).	Enforce statutory lag from target gate: freeze blocks 59-61; revise starting block 62.
EC-07	Multi-Generator Asset Name Mismatch (DynamoDB asset_id not in plant config).	Asset validator catches unrecognized name (e.g. unknown feeder code).	Apply curtailment at plant COMBINED level proportionally across known assets; alert DevOps.
EC-08	Severe Negative DC Drift (Inverter sensor drift outputs negative MW).	Module S1 ADC validation detects $A(b) < 0.00$ MW during daylight.	Clamp $A(b) = 0.00$ MW; exclude tainted block from rolling Performance Ratio calculation.
EC-09	Overcast Solar False Peak (NWP predicts clear-sky but satellite shows cloud).	Guardrail 3 evaluates low $K_t < 0.50$ and high cloud cover > 75%.	Enforce overcast ceiling: cap predictions at $\max(1.10 * P_base, 0.25 * P_CAP_AC)$.
EC-10	Drivetrain Gearbox Ramp Exceedance ($ \Delta P > 15\%$ nameplate capacity).	Guardrail 4 identifies sudden multi-megawatt jump between adjacent blocks.	Smooth transition by clamping step change to maximum allowable ramp rate.
EC-11	Jewli Wind LLM Isolation (Attempted LLM query on Jewli plant).	Module W9 executes <code>_is_llm_disabled_for_plant('JEW LI')</code> .	Bypass LLM query completely; deploy deterministic Phase 6 physics baseline.
EC-12	S3 Data Lake Multi-Part Upload Network Timeout.	boto3 client catches <code>botocore.exceptions.EndpointConnectionError</code> .	Execute 3 exponential retries; fallback to local /tmp caching; log critical S3 alert.

5. Architectural Certification and Governance Sign-off

This document constitutes the complete, approved, and authoritative technical specification for the Intellis Intraday Scheduling Architecture (DOC-M04-S01-005, Version v1.0). Any modification to mathematical formulations, physical guardrails, regulatory freeze lag definitions, or database schemas must be submitted as a formal revision (v1.1+) through the VPPL Engineering Change Control Board (CCB) and recorded in the Master Documentation Registry.